

قوانين للفاينل و الميجرز:

Secant method

False position

Modified newton's

Aitken's method

Steffson method

Muller's method و الفورملا الاساسية لازم تحفظيهم c بس ال a and b بتجيب الفورملا حقت

Polynomial of centered

Polynomial of hermite

Cubic spline b will be given

Closed cotes simpson's 3th/8 and $n=4$

Open cotes all will be given except the midpoint

4.4 all will be given

5.3 the T_n will be given and the formula

rule

Chain rule

quotient rule

Derivative

$$\frac{d}{dx} n = 0$$

$$\frac{d}{dx} x = 1$$

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} n^x = n^x \ln n$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \cot x = -\csc^2 x$$

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\frac{d}{dx} \csc x = -\csc x \cot x$$

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \operatorname{arccot} x = -\frac{1}{1+x^2}$$

$$\frac{d}{dx} \operatorname{arcsec} x = \frac{1}{x\sqrt{x^2-1}}$$

$$\frac{d}{dx} \operatorname{arccsc} x = -\frac{1}{x\sqrt{x^2-1}}$$

Integral (Antiderivative)

$$\int 0 \, dx = C$$

$$\int 1 \, dx = x + C$$

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

$$\int e^x \, dx = e^x + C$$

$$\int \frac{1}{x} \, dx = \ln x + C$$

$$\int n^x \, dx = \frac{n^x}{\ln n} + C$$

$$\int \cos x \, dx = \sin x + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \sec x \, dx = \sec x + C$$

$$\int \cot x \csc x \, dx = -\csc x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$$

$$\int -\frac{1}{\sqrt{1-x^2}} \, dx = \arccos x + C$$

$$\int \frac{1}{1+x^2} \, dx = \arctan x + C$$

$$\int -\frac{1}{1+x^2} \, dx = \operatorname{arccot} x + C$$

$$\int \frac{1}{x\sqrt{x^2-1}} \, dx = \operatorname{arcsec} x + C$$

$$\int -\frac{1}{x\sqrt{x^2-1}} \, dx = \operatorname{arccsc} x + C$$

Jet P2050 series

$$\begin{array}{l|l} \frac{d}{dx}(x) = 1 & \frac{d}{dx}(x^7) = 7x^6 \\ \frac{d}{dx}(5x) = 5 & \frac{d}{dx}(3x^7) = 21x^6 \\ \frac{d}{dx}(x^2) = 2x & \frac{d}{dx}(\ln x) = \frac{1}{x} \\ \frac{d}{dx}(7x^2) = 14x & \frac{d}{dx}(3 \ln x) = 3 \frac{1}{x} = \frac{3}{x} \end{array}$$

$$\begin{array}{l|l} \frac{d}{dx}(e^x) = e^x & \frac{d}{dx}(\sin x) = \cos x \\ \frac{d}{dx}(e^{3x}) = 3e^{3x} & \frac{d}{dx}(\sin 3x) = 3 \cos 3x \\ \frac{d}{dx}(e^{-x}) = -e^{-x} & \frac{d}{dx}(5 \sin 4x) = 20 \cos 4x \\ \frac{d}{dx}(e^{-7x}) = -7e^{-7x} & \frac{d}{dx}(\cos x) = -\sin x \\ \frac{d}{dx}(3e^{-7x}) = -21e^{-7x} & \frac{d}{dx}(3 \cos x) = -3 \sin x \\ & \frac{d}{dx}(2 \cos 5x) = -10 \sin 5x \end{array}$$

$$\begin{aligned} & \frac{d}{dx}(2x \cos 3x) \\ &= 2 \cos 3x + 2x(-3 \sin 3x) \\ f'(x) &= 2 \cos 3x - 6x \sin 3x \\ f''(x) &= -6 \sin 3x - [6 \sin 3x + 6x \cos 3x] \\ &= -6 \sin 3x - 6 \sin 3x - 18x \cos 3x \\ &= -12 \sin 3x - 18x \cos 3x \end{aligned}$$

Product rule % دالین بینیم جزو

$$\frac{d}{dx} [f(x) \cdot g(x)] = f(x)g'(x) + f'(x)g(x)$$

Chain rule % دال داخل دال

$$\frac{d}{dx} [(f(x))^n] = n(f(x))^{n-1} \cdot f'(x)$$

$$\frac{d}{dx} [f(g(x))] = f'(g(x))g'(x)$$

Quotient rule % دال مقسومہ علی دال

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - f(x)g'(x)}{(g(x))^2}$$